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Teaching Statement

Teaching Philosophy

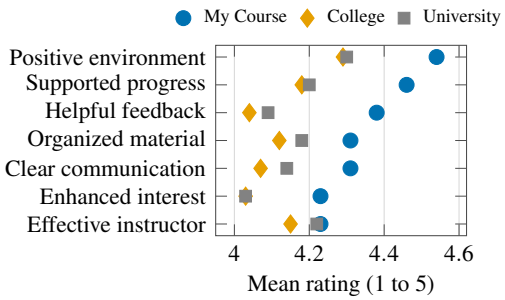
Throughout my academic journey, I have been shaped by educators who did more than teach technical methods; they cultivated curiosity, rigorous reasoning, and clear communication. I strive to carry that legacy forward in my own classroom. In rapidly evolving fields such as cybersecurity and machine learning, memorizing today's tools is never enough; I want students to leave my courses able to dissect an unfamiliar problem, design and execute a sound experiment, and explain why the empirical evidence supports their conclusion. Because students develop that judgment by moving iteratively between conceptual foundations, guided practice, and independent inquiry, my teaching philosophy centers on an interactive, hands-on, and project-based learning environment. My goal is to make complex ideas approachable while empowering students to question assumptions, test systems under realistic constraints, and defend their engineering decisions.

Prior Teaching and Mentoring Experience

My commitment to teaching has been forged across more than six years as an Instructor of Record at three institutions. At Rochester Institute of Technology (RIT), I have served as the primary instructor for both graduate and undergraduate courses: *CSEC 720: Deep Learning Security* (a graduate seminar of 13 M.S. and Ph.D. students connecting foundational machine learning with adversarial robustness and privacy) and *CSEC 140: Introduction to Cybersecurity* (an undergraduate course of approximately 30 students connecting core security principles, cryptography, and network defense to everyday systems). I have also delivered invited guest lectures across graduate cybersecurity courses at RIT.

Prior to and during the early stages of my doctoral studies, I served as a full-time faculty member in Bangladesh—first as a Lecturer at North Western University (NWU) and subsequently as a Lecturer and Assistant Professor in Computer Science and Engineering at Khulna University of Engineering & Technology (KUET). Across both universities, I designed lectures, laboratories, assessments, and term projects for courses spanning *Machine Learning*, *Computer Security*, *Computer Networks*, *Software Engineering*, *Mobile Computing*, and *Algorithm Analysis & Design*, managing class sizes of approximately 40 students at NWU and 120 students across multiple sections at KUET. Additionally, I taught machine learning and computer networks to K–12 educators at the Government Teachers' Training Institute in Khulna, an experience that sharpened my ability to adapt technical concepts to learners with diverse backgrounds and career goals.

Beyond classroom instruction, mentoring student researchers has been one of the most rewarding aspects of my academic career. At RIT, I have mentored 26 students across degree levels (5 junior Ph.D., 4 M.S., and 17 B.S. students) in adversarial machine learning, trustworthy generative AI, and media forensics. At KUET, I advised more than 30 undergraduate thesis and capstone students, resulting in seven student-led peer-reviewed publications.



> *"Akib did a great job communicating and explaining the different machine-learning algorithms from the beginning of the semester. The structure of the class was also set up very well. He was always eager to check in with students to answer any questions."*
>
> Anonymous student, CSEC 720 SmartEvals

> *"I appreciate his high flexibility and accessibility. He also presented the content and facilitated the discussions very well."*
>
> Anonymous student, CSEC 720 SmartEvals

Figure 1: CSEC 720 student evaluations (Spring 2025; 13 of 13 students responded) alongside representative anonymous comments.

Student Feedback and Continuous Improvement. In the Spring 2025 evaluations for *CSEC 720: Deep Learning Security* (100% response rate, 13/13 students), my ratings exceeded both College and University averages across every evaluated dimension (Figure 1). The largest positive margins occurred in providing helpful feedback (4.38 vs. 4.04 college mean), supporting student progress (4.46 vs. 4.18), and fostering a positive learning environment (4.54 vs. 4.29). Building on constructive student suggestions from this offering, I have also introduced intermediate grading checkpoints and cumulative grade-tracking rubrics so students receive rapid, actionable feedback before each subsequent project milestone.

Approach to Teaching

Active Learning and Critical Discussion. Rather than relying on passive lectures, I introduce a core concept through a motivating real-world failure mode and pose a structured problem for students to tackle individually or in small groups before reconvening for class-wide synthesis. In graduate seminars, students annotate assigned papers on rotating activity sheets, responding directly to peers' critiques. Student presentations are evaluated not only on delivery, but also on the depth of questions they pose to classmates, training them to progress from simple clarifying questions to threat-model challenges (e.g., asking how a defense behaves when an adversary's query budget or access changes). Structured around Bloom's taxonomy, these exercises surface misconceptions early, broaden classroom participation, and allow me to adapt pacing in real time, supplemented by an active course messaging channel for shared clarifications.

Hands-On Experimentation. Cybersecurity and machine learning are empirical disciplines best mastered when students can test theoretical claims against observable system behavior. To keep students focused on core scientific decisions rather than incidental environment setup, I provide clean, runnable baseline pipelines and ask students to modify a critical mechanism, justify their design choice, and analyze where the method succeeds or breaks. For example, in an adversarial machine learning lab, students craft constrained evasion attacks, vary perturbation budgets, and test whether adversarial training against one attack family transfers to unseen threat models. Across both RIT and KUET laboratories, writing functional code is only the first step: students must articulate what their experiment proves, what confounds remain, and how a peer can reproduce their results.

Project-Based Learning and Societal Impact. Semester-long projects empower students to transition from solving predefined labs to formulating and defending an original research question. I scaffold this journey through five milestones: a literature review, a problem proposal, mid-semester check-ins, structured peer review, and a final conference-style paper, presentation, and reproducible codebase. In one-on-one advising meetings, I help students scope ambitious ideas into tractable experiments while preserving their intellectual ownership; recent student projects have explored NeRF-based deepfake forensics, poisoning attacks on language models, and LLM-driven honeypots. Throughout both labs and projects, I challenge students to connect technical metrics to broader societal impact, drawing on my experience with the NSF VARUNA and DeFake projects to illustrate how false positives, false negatives, and threat-model assumptions directly affect human privacy, evidentiary integrity, and public trust.

Clear Expectations and Inclusive Support. Students are most likely to meet high standards when expectations are clear and appropriate support is readily available. Because students enter my courses with varied programming and research backgrounds, I provide modular starter code, explicit deliverables, and rubrics that evaluate implementation, empirical analysis, and written communication separately. To support student well-being without compromising rigor, I incorporate structured flexibility (such as "Oops Tokens" for penalty-free deadline extensions) alongside a transparent generative-AI policy that permits AI assistance for brainstorming and debugging while requiring full attribution and independent verification.

Approach to Mentorship

I view research mentorship as a structured, gradual transfer of intellectual ownership. When onboarding new students, I work side by side with them to curate foundational papers, formulate a crisp hypothesis, and build a minimal working prototype. As their confidence grows, I challenge them to propose the next experimental control and predict what competing outcomes would imply, steadily handing over responsibility for project direction. Equally important, I also teach the unspoken skills of research: reading papers critically, keeping code reproducible, writing clear rebuttals, presenting at conferences, and preparing for academic or industry careers. Having studied and taught in both Bangladesh and the United States, I am deeply attentive to how cultural and

educational backgrounds shape student expectations. In my own research group, I will set clear expectations for meetings, feedback, and communication, encourage students to raise difficulties early, and connect them with university resources when they need support.

Supporting students holistically also means building community and bridging academia with industry. As **Vice President of RIT's Doctoral Student Association (2022–2025)**, I organized research-skills workshops, peer mentoring events, and alumni career panels; at KUET, I organized hackathons and programming contests that connected students with industry practitioners. Drawing on my industry experience, from leading an edge-AI engineering team at SmartRoad.AI to serving as a Principal ML Architecture Consultant at Apprendi AI and Technical Strategy Advisor at Zatiq, I will regularly host industry researchers in our lab meetings and connect students with external internships and cross-institutional collaborations, ensuring they graduate prepared to lead in either academia or industry.

Courses and Teaching Plan

I am prepared to teach a wide range of undergraduate and graduate courses in **cybersecurity and privacy** (e.g., *Introduction to Cybersecurity*, *Computer and Network Security*, *Privacy-Enhancing Technologies*), **machine learning and computer vision** (e.g., *Deep Learning Security*, *Machine Learning*, *Computer Vision*), and **core computer science** (*Algorithms*, *Software Engineering*, *Computer Networks*, and *Mobile Computing*). I adjust the balance between guided instruction and independent inquiry according to the course level. Introductory courses emphasize conceptual clarity and guided laboratory practice, whereas advanced courses give students greater responsibility for developing threat models, evaluating empirical evidence, and designing independent research.

In addition to core offerings, I am excited to develop new graduate research seminars at the intersection of my research program, specifically **Secure Agentic AI** and **Synthetic Media Forensics** (incorporating foundations of autonomous LLM agents, adversarial machine learning, and trustworthy generative AI). Structured around critical paper reviews, reproducible red-teaming/forensic benchmarks, and an original research project, these seminars will equip students to tackle emerging security challenges while serving as a fertile incubator for collaborative student research.